

BIGRED//HACKS 2026 • THEME: NAVIGATION

CLYF

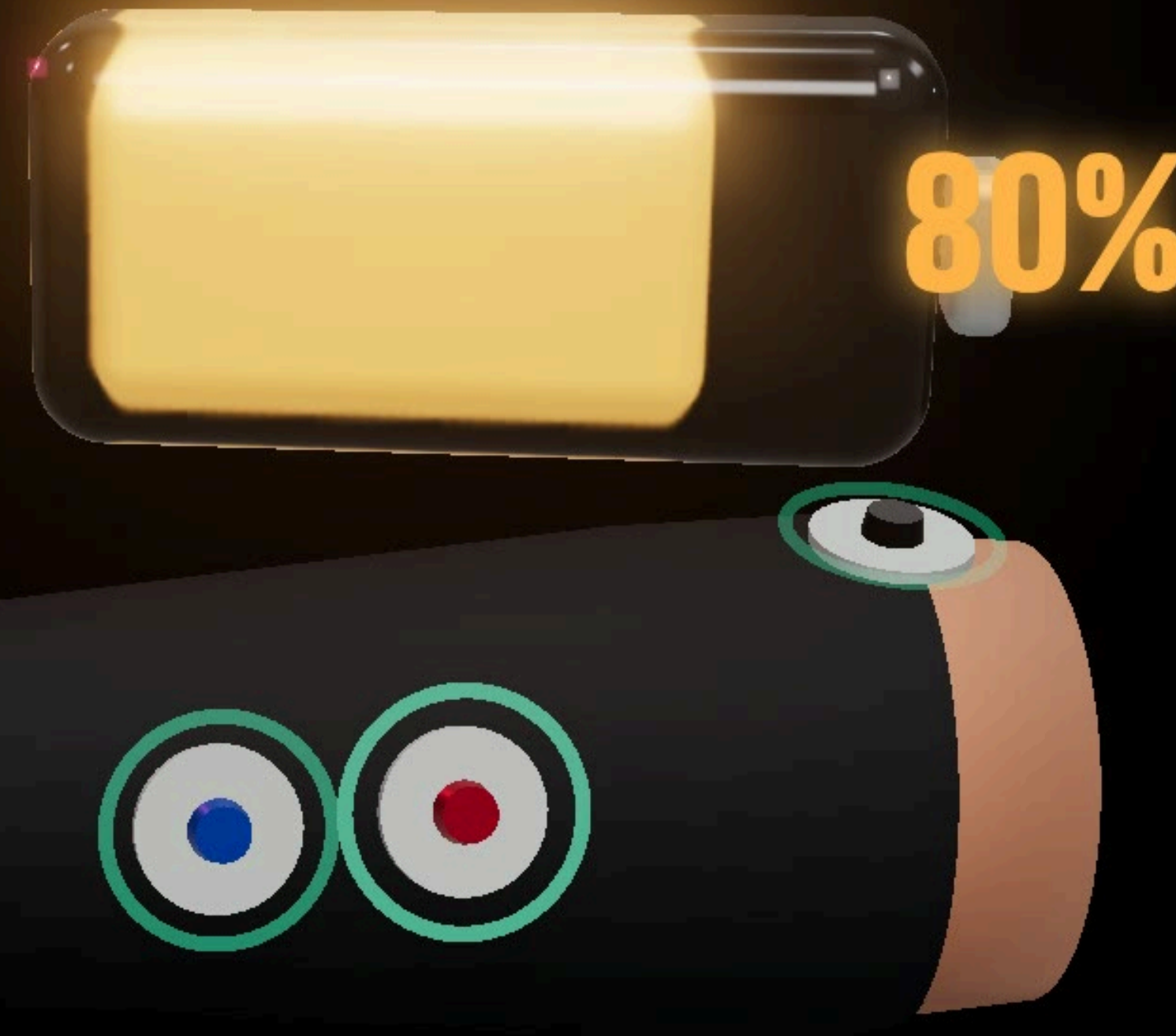
A forearm sleeve that tells climbers
when to climb again.

TEAM

Jeremy Li

BUILT WITH

Arduino UNO Q • MyoWare 2.0 sEMG



THE PROBLEM

CLIMBERS GUESS HOW LONG TO REST.

THE PUMP

On a hard route the forearms give out first. Grip fades, and the climb ends.

THE SHAKE-OUT

Hanging off a jug mid-route, you rest with no idea when you have actually recovered.

THE COST

Rest too little and you fall off the next move. Rest too long and you burn the session.

Maps give drivers an ETA. Climbers get **guesswork**.

OUR SOLUTION

A FOREARM BATTERY WITH AN ETA.

● GRIPPING

Effort above a sustainable level drains the battery.

● RESTING

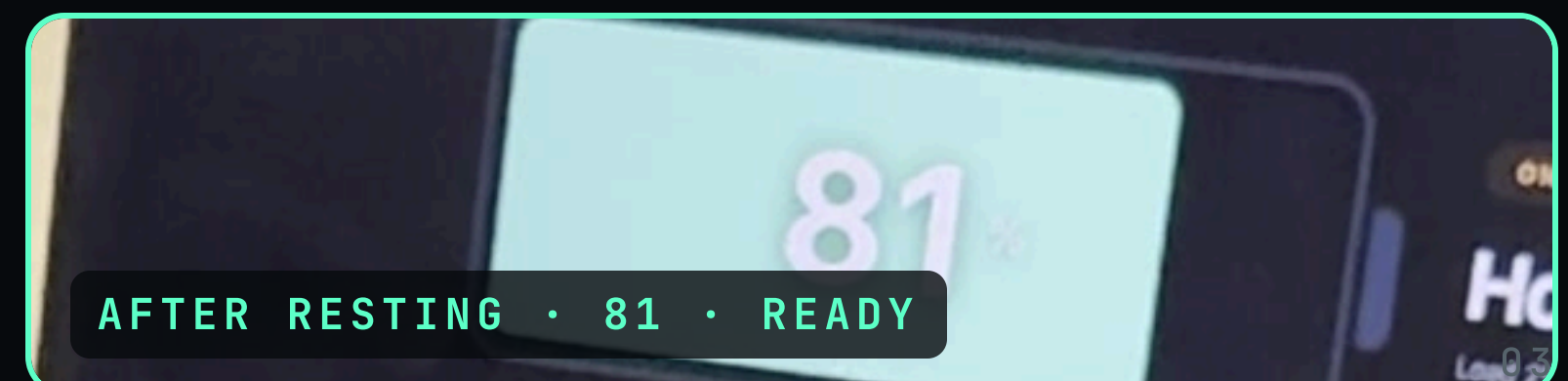
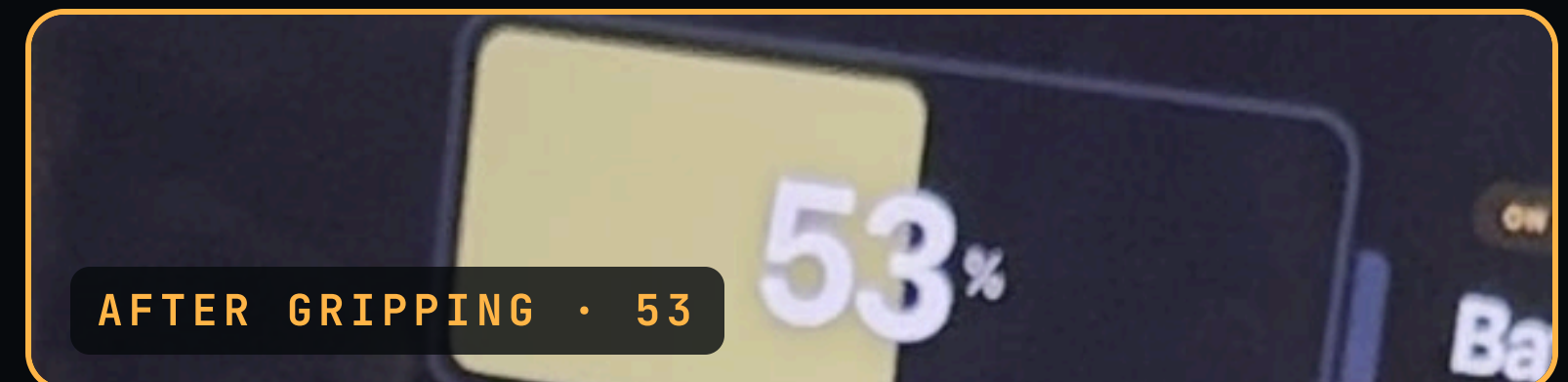
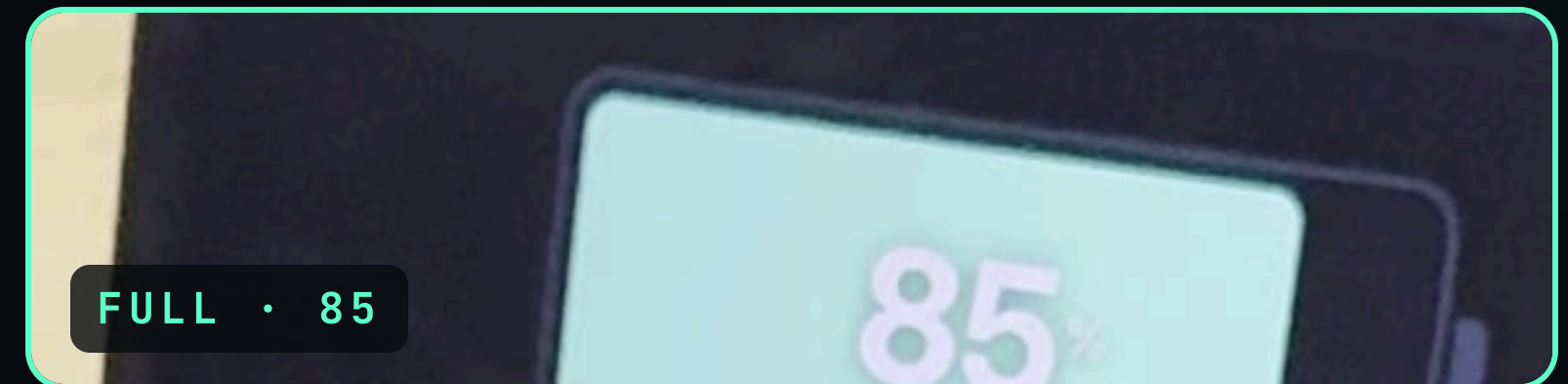
It recharges, with a countdown: **“Ready in 12 s”**.

● RELAX FULLY

EMG sees a tense forearm and slows the recharge.

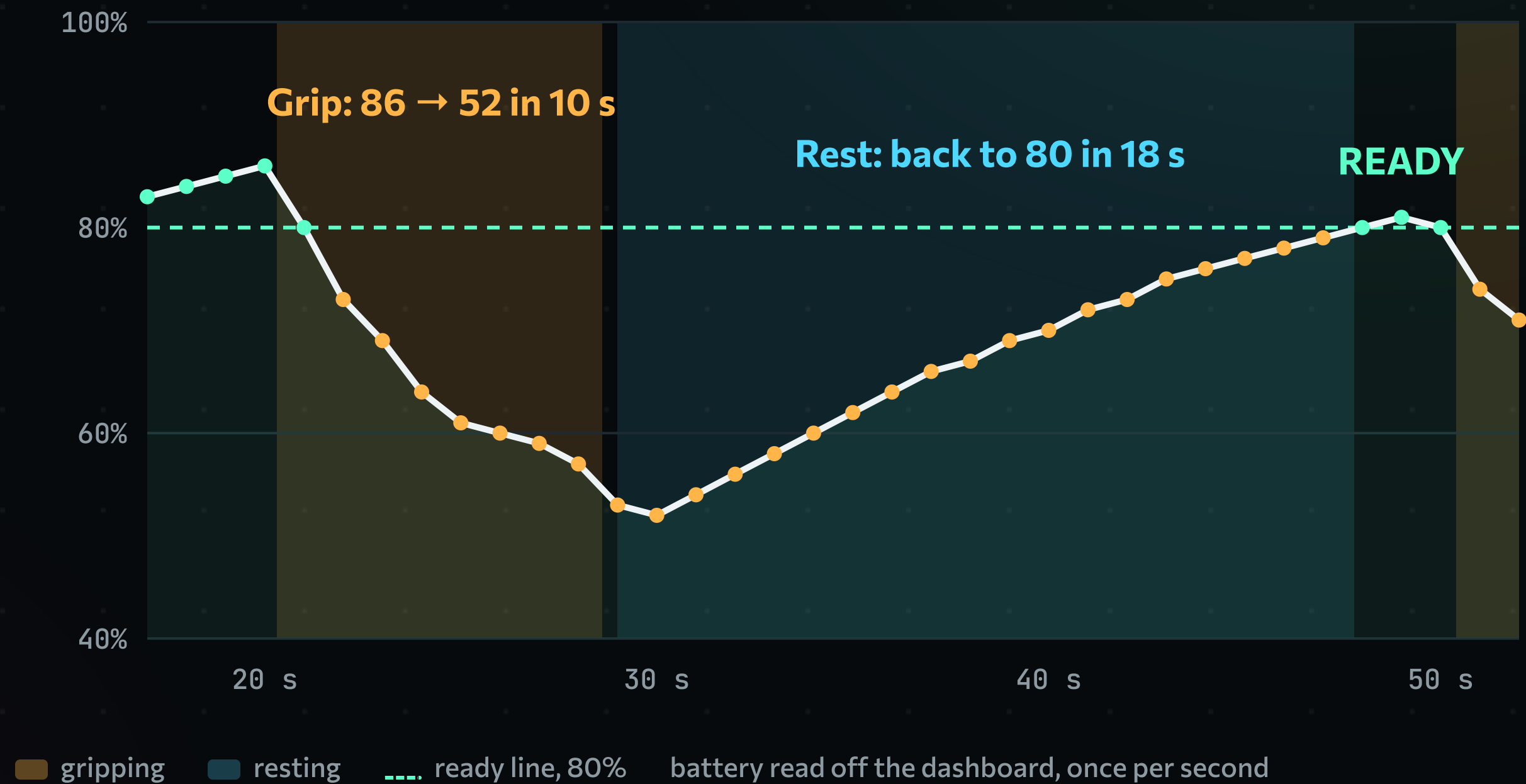
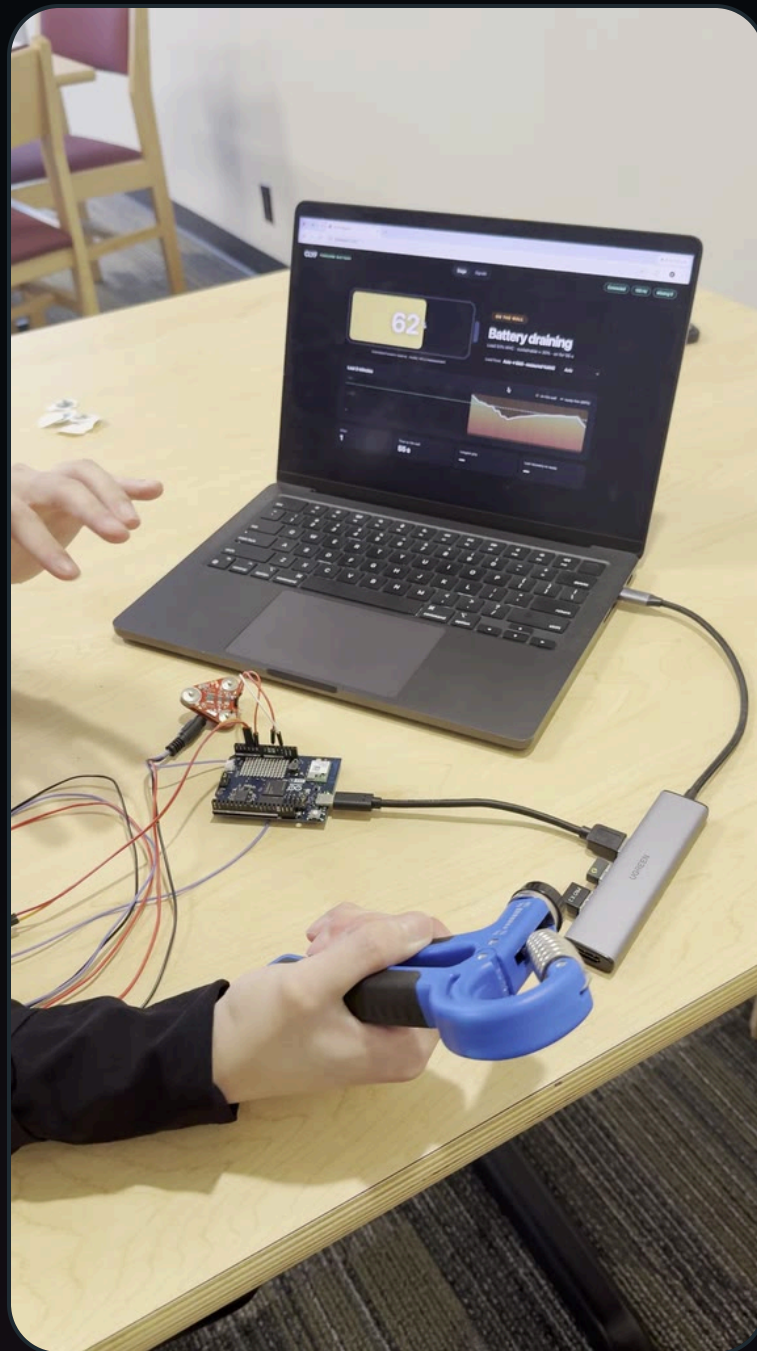
● READY

At 80% it glows and chimes: **climb on.**



DEMO • ONE REAL SESSION

GRIP DRAINS IT. REST BRINGS IT BACK.



HOW WE BUILT IT

FROM MUSCLE TO MINUTES.

01 · SENSE

THE SLEEVE

Surface EMG on the finger flexors, plus a force sensor on the hold.

MyoWare 2.0 + cable shield
FSR, 4 mm

02 · SAMPLE

ARDUINO UNO Q

The STM32 reads both channels on a fixed clock and streams every sample.

100 Hz · 12-bit ADC
0 dropped samples

03 · CLEAN

LINUX SIDE

Python on the board relays to the laptop over USB. No Wi-Fi needed.

Bridge RPC · Socket.IO
ADB port forward

04 · NAVIGATE

DASHBOARD

Auto-calibrated %MVC, bad-signal detection, and the battery model.

JavaScript · Canvas
ready at 80% · $\tau = 20$ s

Arduino UNO Q

STM32 sketch

MyoWare 2.0

FSR 400

Python

Socket.IO

JavaScript

HTML5 Canvas

ADB

three.js (video)

WHAT TO TRUST

MEASURED, THEN MODELLED.

MEASURED

- Muscle effort, as % of your own max grip
- Hand on the hold
- How long each grip and rest lasted
- How relaxed the forearm is while resting

ESTIMATED

- Battery level and “ready in” countdown, from a W'-balance style model
- Labelled as an estimate on screen
- Demo constants, not yet fitted per climber

Spiky EMG

Calibrating from peaks made hard grips read 30–40%. Per-second averages fixed it.

Loose electrodes

A pinned or flat signal is flagged, not turned into a number.

Campus Wi-Fi

Device isolation blocked the link, so data runs over USB.

No resistor

The FSR floats without one, so it works as a contact switch for now.

IMPACT

TURN A FEELING INTO A NUMBER.

→ CLIMBERS

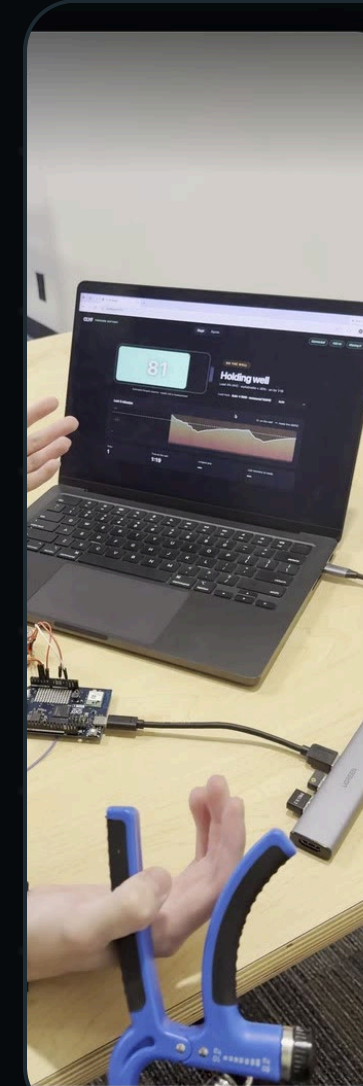
Rest exactly as long as you need on a project, and see effort across a session.

→ GYMS AND COACHES

Objective grip load per attempt instead of “how pumped were you?”

→ BEYOND CLIMBING

Any grip-heavy activity where recovery decides performance.



LIVE DASHBOARD

• READY • CLIMB



Estimated forearm reserve - zoomed from the footage

Okay, now it's back, so you should be climbing again.

WHAT'S NEXT

EYES ON THE WALL, NOT ON A SCREEN.

1 READY ON THE WRIST

A haptic tap from the watch when you're recovered. The Bluetooth link is already built.

2 FATIGUE YOU CAN SEE

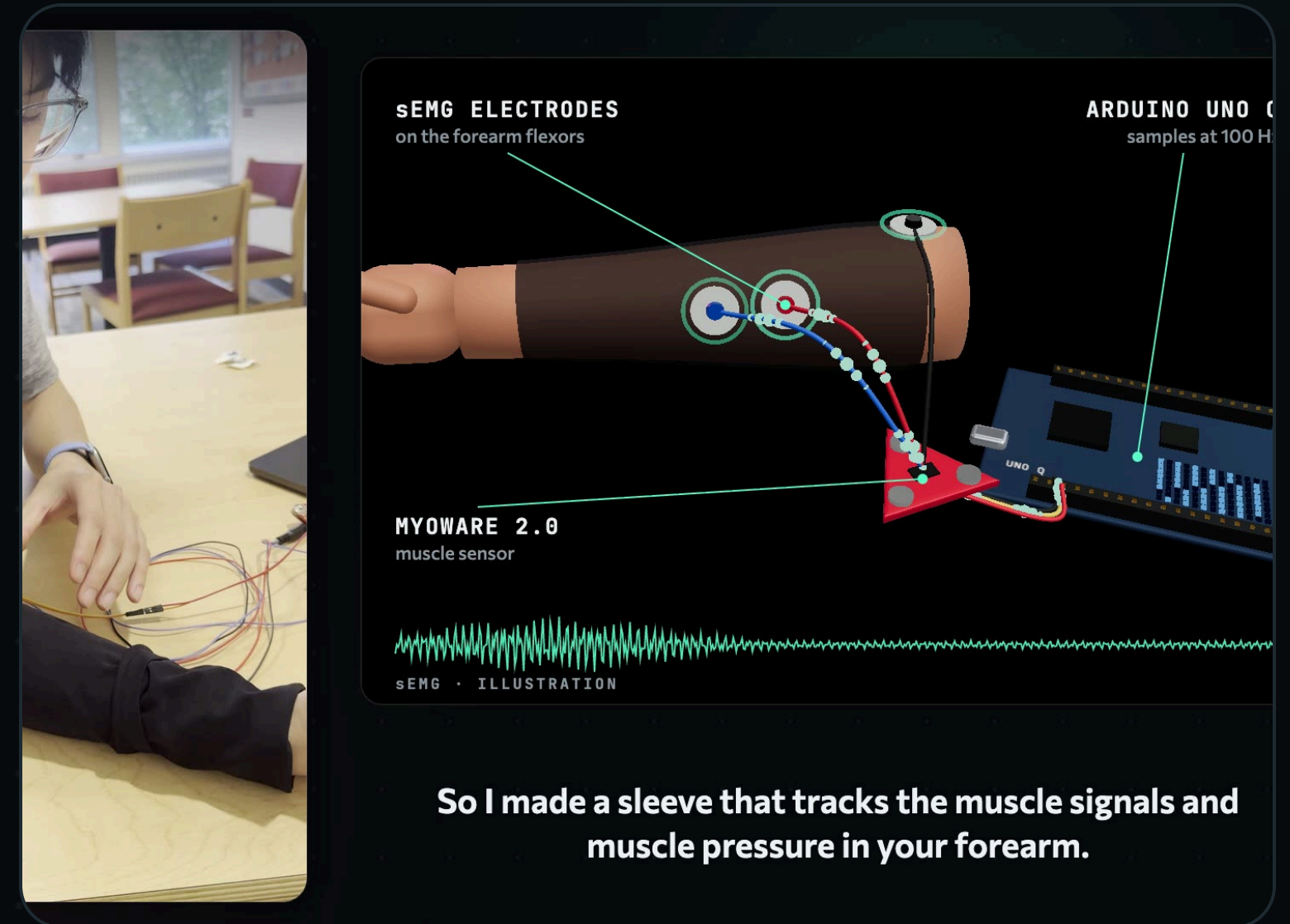
Track the raw EMG's frequency shift, a known marker of muscle fatigue.

3 FITTED TO YOU

Learn each climber's own drain and recovery rates from their sessions.

4 WIRELESS SLEEVE

Smaller board, battery power, and a cleaner fit.



THANK YOU
CLYF

Know when to **climb again.**

Jeremy Li · BigRed//Hacks 2026